Review Articles
Systematic Reviews

Comparison of Autograft Types in Anterior Cruciate Ligament Reconstruction

A Systematic Review and Bayesian Network Meta-Analysis of Randomized Clinical Trials

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Background: The literature regarding optimal autograft choice for anterior cruciate ligament (ACL) reconstruction (ACLR) remains inconclusive. This network meta-analysis (NMA) compares common autografts for primary ACLR.

Methods: PubMed, Scopus, Web of Science, and Embase were searched up to May 3, 2025, for randomized clinical trials (RCTs) on primary ACLR in adults that compared ≥2 of the following tendon autografts: 4-strand semitendinosus (4SST), 4-strand semitendinosus-gracilis (4SSTG), its 5-strand variant (5SSTG), bone-patellar tendon-bone (BPTB), quadriceps tendon with bone (QTB), and free quadriceps tendon (FQT). Outcomes analyzed in the NMA were the International Knee Documentation Committee (IKDC) subjective score, Lysholm score, Tegner Activity Scale, anteroposterior (instrumented) and rotational (pivot-shift) stability, and rerupture or revision ACLR rate. Autografts were ranked using surface under the cumulative ranking (SUCRA) values.

Results: A total of 44 RCTs with 3,491 patients were included in the NMA. With respect to the IKDC, QTB was statistically superior to BPTB (mean difference = 3.46, 95% credible interval [CrI]: 0.29 to 6.77), although the difference was likely not clinically meaningful. QTB ranked highest for the IKDC (SUCRA = 90.1%) and Tegner (SUCRA = 85.3%), while BPTB ranked lowest for the IKDC and Lysholm. With respect to knee laxity, QTB ranked second in anteroposterior and first in rotational stability, and it carried a significantly lower risk of a 2+ or higher pivot-shift than 4SST (risk ratio = 0.26, 95% CrI: 0.07 to 0.85). QTB was associated with a decreased risk of rerupture/revision compared with other autografts (SUCRA = 83.3%).

Conclusions: Based on the autograft rankings, QTB was found to lead to improved functional, activity-related, and stability outcomes overall, while also reducing the risk of graft failure.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Anterior cruciate ligament (ACL) injuries are common orthopaedic sports injuries and a significant global health-care burden[1,2], often leading to secondary injuries, long-term instability, and osteoarthritis[3-5]. While ACL reconstruction (ACLR) using autografts is a standard treatment, the optimal choice of autograft, typically among hamstring tendon (HT), bone-patellar tendon-bone (BPTB), and quadriceps tendon (QT), remains controversial[6,7]. Conventional studies on this topic have often relied on pairwise meta-analyses, which are limited in their ability to account for the complexity of multiple autograft comparisons[8-11]. Subsequent network meta-analyses (NMAs) attempted to address this issue.

To our knowledge, 7 previous NMAs have addressed graft selection for ACLR[12-18]. However, their findings were limited by significant methodological inconsistencies, including the inclusion of heterogeneous graft types, such as allografts, synthetic grafts, and autografts, in 1 network[15-18], an approach that might provide a broad overview but does not yield a clear consensus, as these graft types differ substantially in their clinical indications. Given that autografts remain the preferred

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choice for ACLR[19-22], conducting an NMA dedicated to their comparison using standardized graft classifications is essential. Previous NMAs have often misclassified autograft types, by grouping different HT variants together[12,13,16], not specifying bone block use in patellar tendon autografts[13,15,16], and either omitting QT autografts[14,16,18] or not distinguishing between QT variants with versus without bone[12,13,15].

This systematic review and Bayesian NMA aimed to address these gaps by comparing 6 well-defined autograft subtypes for primary ACLR in adults with respect to patient-reported outcome measures (PROMs), knee stability, graft failure, and return to sport. We hypothesized that there would be no significant differences among the compared autografts with respect to these outcomes.

> *QT autografts performed the best overall with respect to the PROMs… QTB [QT with bone] was the most favorable autograft with respect to the risk of ACL rerupture or revision ACLR.*

Materials and Methods

This systematic review and NMA was performed according to the PRISMA-NMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension Statement for Reporting of Systematic Reviews Incorporating Network Meta-Analyses of Health Care Interventions)[23], and the protocol was registered in PROSPERO (CRD420251117682).

Search Strategy

An electronic search was conducted in PubMed, Scopus, Web of Science, and Embase, from inception to May 3, 2025, without language restrictions. The search strategy was developed using keywords and MeSH terms relating to ACLR, different autograft types, and randomized clinical trials (RCTs). The detailed search strategy is available in Appendix 1.

Eligibility Criteria and Outcome Definition

We included RCTs comparing ≥2 autografts for primary ACLR in adults, with eligibility criteria detailed in Table 1. Included studies had to report data on at least 1 predefined outcome. These included PROMs, namely the International Knee Documentation Committee (IKDC) Subjective Knee Form[24], Lysholm score[25], and Tegner Activity Scale[25]. Additional outcomes involved knee stability measures, namely instrumented anteroposterior (AP) laxity and the pivot-shift test result. The other outcomes were ACL rerupture or the need for revision surgery and return to sport. For homogeneity, studies reporting AP laxity data were limited to those employing a standardized 134 N or maximum manual force. For RCTs with multiple published reports, the one with the most recent follow-up was used; however, if an outcome was missing from that report, the next latest report was included.

Study Selection and Data Extraction Process

Three authors (S.Y., S.M.M., and F.S.) independently performed title and abstract screening, followed by full-text screening of relevant studies. A senior author (F.V.) was consulted to resolve any disagreements. Data extracted included study details (title, first author, year, country, follow-up, design), patient characteristics (sample size, age, sex, body mass index [BMI], time from injury to surgery), inclusion and exclusion criteria, athlete level (general, professional), graft details (type, subtype, diameter, fixation), and outcome data from the latest follow-up.

Risk-of-Bias Assessment

Risk of bias was assessed using the Cochrane risk-of-bias tool (RoB 2) for randomized trials[26]. RoB plots were generated using the robvis web app (Luke McGuinness)[27].

Statistical Analysis

We performed an NMA within a Bayesian framework using Markov-chain Monte Carlo (MCMC) simulation. Model convergence was assessed using trace plots and the Gelman-Rubin diagnostic. Treatment effects are reported as the mean difference (MD) for continuous outcomes and as the risk ratio (RR) for binary outcomes, with the 95% credible interval (CrI). When the effect estimate for a continuous outcome reached significance, we further assessed whether it exceeded the minimal clinically important difference (MCID). Publication bias and small-study effects were assessed using funnel plots and the Egger test. Treatment rankings were estimated using surface under the cumulative ranking (SUCRA) values. Full, detailed descriptions of our statistical analysis methods are available in Appendix 2.

Results

Study Selection and Characteristics of the Included Studies

The study selection process is illustrated in Figure 1. A total of 48 reports representing 45 RCTs were included in the systematic review[28-75], and 44 of the 45 were ultimately included in the NMA. These reports included 3,491 patients and were from 25 countries. For 3 RCTs, data were extracted from 2 separate reports, as different follow-up reports provided data on different outcomes[29,52,55,58,61,70]. Characteristics of the included studies are summarized in Table 2 and in Appendix 3.

Network Geometry

Network characteristics and plots for all outcomes are provided in Appendix 4, including Supplementary Figure 1. The

Table 1. Inclusion and Exclusion Criteria*

Inclusion Criteria	Exclusion Criteria
Population	Population
Patients aged 18 years or older Patients diagnosed with ACL rupture	Patients with a history of ACL rupture Patients with a history of relevant chronic diseases
Intervention	Intervention
Primary ACL reconstruction	Revision ACL reconstruction Any ACL repair technique ACL remnant preservation technique Graft augmentation technique Double-bundle reconstruction
Comparison	Comparison
Studies comparing at least 2 of the specified graft types: Quadrupled (4-strand) semitendinosus, abbreviated as *4SST* Double-looped (4-strand) semitendinosus and gracilis, abbreviated as *4SSTG* 5-strand hamstring graft consisting of both semitendinosus and gracilis, abbreviated as *5SSTG* Bone-patellar tendon-bone graft with bone blocks on both ends, abbreviated as *BPTB* All-soft-tissue (free) quadriceps tendon, abbreviated as *FQT* Quadriceps tendon with patellar bone block, abbreviated as *QTB*	Studies involving none of the specified graft types Studies that do not compare at least 2 graft types of interest Studies that do not specify which hamstring tendons were used for the graft Studies that do not report the number of strands used in the hamstring graft Studies that do not specify the number of bone block ends in the patellar graft Studies involving allografts Studies involving artificial grafts
Outcome	Outcome
Studies reporting at least 1 of the following outcomes: IKDC Lysholm Tegner Instrumented laxity Pivot-shift test Rerupture or revision ACL reconstruction Return to sport	Studies with insufficient outcome data Studies reporting none of the specified outcomes
Study design	Study design
Randomized clinical trials	Reports with shorter follow-up reporting data on the same outcomes as a publication with longer follow-up Nonrandomized clinical trials Observational studies, including case reports, case series, cohort studies, and case-control studies Conference abstracts, pilot studies, commentaries

*IKDC = International Knee Documentation Committee Subjective Knee Form.

pivot-shift analysis lacked data for 1 of the autografts, QT without bone (free QT [FQT]). All 6 autografts were included in the analyses of the remaining outcomes.

Risk of Bias

RoB assessment is provided in Appendix 5, including Supplementary Figures 2-A and 2-B. Of the 48 included reports, 35 were deemed to have a low risk of bias, 7 exhibited some concerns regarding risk of bias[30,48,50,54,55,58,67], and 6 were judged to have a high risk of bias[46,61,62,64-66].

Network Meta-Analysis

Network estimates for pairwise comparisons are provided in Table 3 and shown in forest plots in Appendix 6, including Supplementary Figures 3-A through 3-F. Additionally, SUCRA plots illustrating the ranking probabilities of the autografts for each outcome are shown in Figure 2.

IKDC Subjective Knee Evaluation

The IKDC NMA included 22 studies with a total of 1,419 patients[28,29,31,33,35-41,43-46,48,50,51,54,57,70,72]. The largest MD was observed between quadriceps tendon with bone block (QTB) and BPTB (MD = 3.46 in favor of QTB, 95% CrI: 0.29 to 6.77), which was statistically significant but likely not clinically meaningful. Other pairwise comparisons did not reveal a significant difference. As shown in Figure 2-A, QTB (SUCRA = 90.1%) and 4-strand semitendinosus (4SST) (SUCRA = 70.7%) demonstrated the highest probability of being the most favorable autograft, followed by 5-strand semitendinosus-gracilis (5SSTG) (SUCRA = 45.0%), 4-strand semitendinosus-

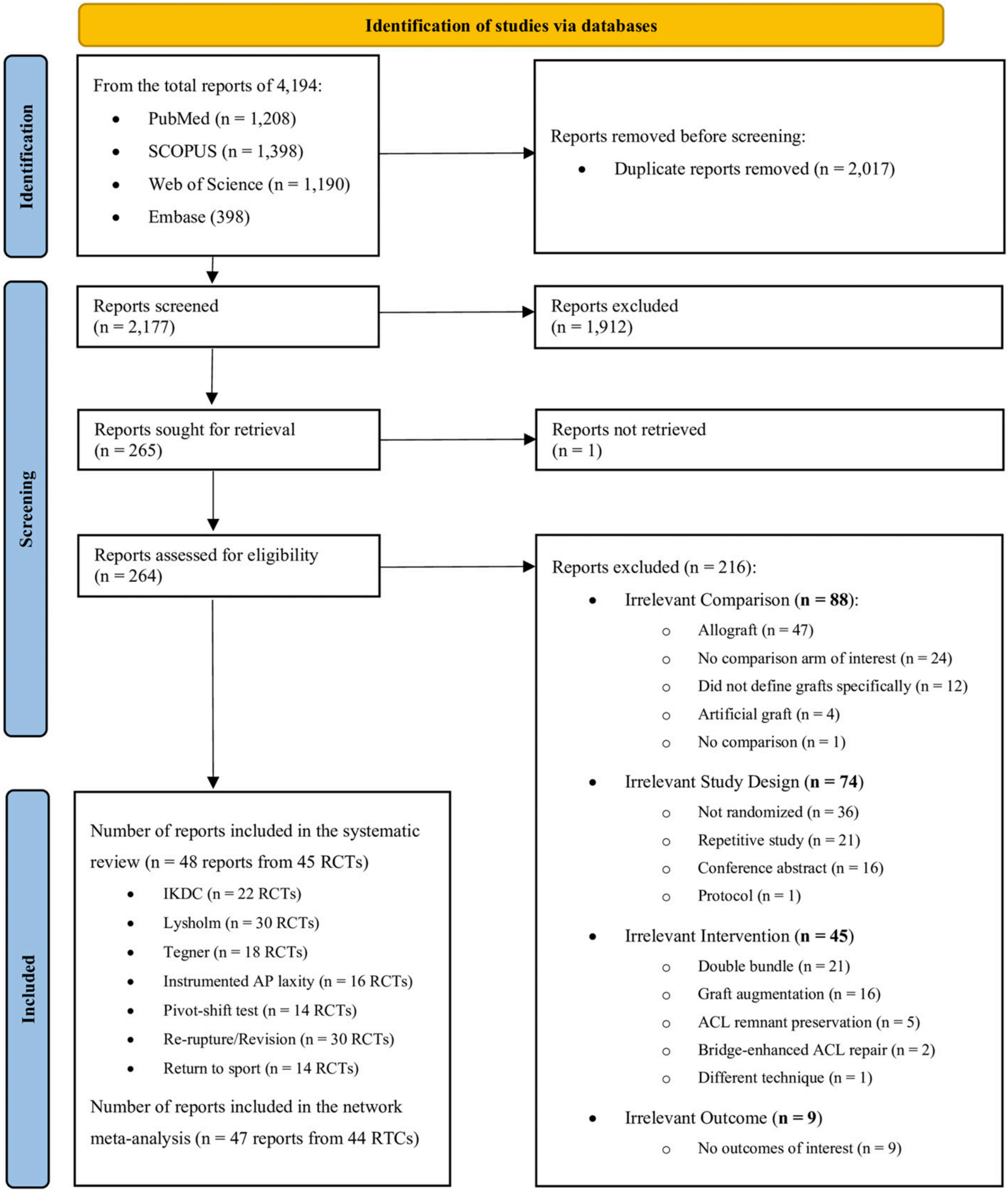

Fig. 1

PRISMA flowchart of the included randomized clinical trials (RCTs) in the systematic review and network meta-analysis.

Table 2. Characteristics of the Included Randomized Clinical Trials*

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age *(yr)*	Sex, F/M
García-Linage, 2025, Mexico[30]	1 yr	Lysholm Tegner Return to sport	FQT	13	26.2 ± 6.2	7/6†
			4SST	13	28.5 ± 7.2	6/7†
Lucidi, 2025, Italy[29]‡	20 yr	IKDC Lysholm Tegner Rerupture/revision	BPTB	19	29.3 ± 7.2	6/13
			4SSTG	20	31.3 ± 7.6	7/13
Solanki, 2025, India[28]§	2 yr	IKDC Lysholm Return to sport	FQT	30	Overall numbers: 20 to 25 yr: 22; 26 to 30 yr: 16; 31 to 35 yr: 14; 36 to 40 yr: 8#	Overall: 20/40
			5SSTG	30		
Ebert, 2024, Australia[35]	2 yr	IKDC Lysholm Tegner Instrumented laxity Rerupture/revision Return to sport	FQT	48	28.1 ± 8.2	29/28
			4SST**	49	29.4 ± 7.7	27/28
Mo, 2024, Norway[33]	2 yr	IKDC Instrumented laxity Pivot-shift Rerupture/revision	4SSTG	44	34.1 ± 10.1	20/29
			4SST	45	34.2 ± 9.5	26/22
Fallah, 2024, Iran[34]	1 yr	Lysholm Rerupture/revision	QTB	25	25.2 ± 3.9	0/25
			BPTB	25	26.3 ± 4.1	0/25
			4SSTG	25	25.8 ± 3.9	0/25
Popovic, 2024, Norway[32]	18 yr	Lysholm Tegner Instrumented laxity Pivot-shift Rerupture/revision	BPTB	47	Overall median (range): 27 (18 to 49)	19/28†
			4SSTG	49		14/35†
Khatri, 2023, India[37]§	2 yr	IKDC Lysholm Return to sport	FQT	17	NS	Overall: 7/27
			5SSTG	17	NS	
Kuliński, 2023, Poland[36]	4.5 yr	IKDC Lysholm Tegner Pivot-shift Rerupture/revision	4SSTG	68	Median (range): 31 (18 to 45)	15/64
			4SST	72	Median (range): 31 (18 to 44)	18/57
Tang, 2024, Turkey[31]	2 yr	IKDC Lysholm	FQT	17	28.1 ± 6.2	0/17†
			4SSTG	16	28.3 ± 8.6	3/13†
Horstmann, 2022, Germany[39]	2 yr	IKDC Lysholm Instrumented laxity Rerupture/revision Return to sport	QTB	23	24.1 ± 3.6	3/21
			4SSTG	26	32.7 ± 11.4	15/12
Komzák, 2022, Czech Republic[38]	28 mo	IKDC Lysholm	QTB	40	29.8 ± NS	19/21
			BPTB	40	31.0 ± NS	17/23
Arida, 2021, Greece[43]	1 yr	IKDC Lysholm Tegner Rerupture/revision	BPTB	29	29.8 ± 11.1	7/23
			4SSTG	24	30.0 ± 11.7	12/18

continued

Table 2. (*continued*)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age *(yr)*	Sex, F/M
Guglielmetti, 2021, Brazil[41]	2 yr	IKDC Lysholm Pivot-shift Rerupture/revision Return to sport	BPTB 4SSTG	31 31	25.2 ± 5.5 24.6 ± 5.3	8/23 12/19
Krishna, 2021, Singapore[40]	2 yr	IKDC Lysholm Tegner Instrumented laxity Pivot-shift Rerupture/revision	4SSTG 5SSTG	28 28	27.6 ± 7.3 26.3 ± 7.3	8/20 5/23
Vilchez-Cavazos, 2020, Mexico[44]	1 yr	IKDC Lysholm	QTB	14	Mean (IQR): 23 (19.5 to 30.5)	2/11††
			4SSTG‡‡	14	Mean (IQR): 23 (20 to 30)	3/12††
Cristiani, 2021, Sweden[42]	2 yr	Rerupture/revision	BPTB 4SSTG	56 50	28.9 ± 5.9 28.4 ± 6.3	21/59 24/56
Smith, 2020, U.S.A.[45]	2 yr	IKDC Instrumented laxity Rerupture/revision Return to sport	BPTB 4SST	32 27	17.8 ± 2.3 17.6 ± 2.5	17/15 19/13
Roger, 2020, France[46]	2 yr	IKDC Instrumented laxity Pivot-shift Rerupture/revision Return to sport	4SSTG 4SST	27 33	30.3 ± 8.5 30.5 ± 8.9	4/23† 7/26†
Rajput, 2020, Pakistan[47]	2 yr	Lysholm	BPTB 4SSTG	22 19	37.5 ± 11.2 37.0 ± 12.4	0/22 0/19
Barié, 2020, Germany[48]	10 yr	IKDC Lysholm Tegner Instrumented laxity Rerupture/revision	QTB BPTB	21 22	30.5 ± 7.8 30.6 ± 7.5	13/17 13/17
Lind, 2020, Denmark[72]	2 yr	IKDC Tegner Instrumented laxity Pivot-shift Rerupture/revision	QTB 4SSTG	44 47	27.2 ± 6.4 27.1 ± 6.1	21/29 24/25
Mohtadi, 2019, Canada[70]§§	5 yr	IKDC Pivot-shift Rerupture/revision	BPTB 4SSTG	103 105	33.8 ± 9.8† 33.7 ± 10.0†	43/60† 47/58†
Gupta, 2019, India[74]	1 yr	Lysholm Rerupture/revision	BPTB 4SSTG	20 20	25.9 ± 6.2# 25.8 ± 7.5#	NS NS
Gupta, 2020, India[73]	2 yr	Lysholm Rerupture/revision Return to sport	BPTB 4SSTG	80 80	25.0 ± 5.8# 24.8 ± 5.0#	1/79# 2/78#
Martin-Alguacil, 2018, Spain[71]	2 yr	Rerupture/revision	QTB 5SSTG	19 18	18.7 ± 3.6 19.2 ± 3.6	3/23 9/16
Sajovic, 2018, Slovenia[69]	17 yr	Lysholm Instrumented laxity Rerupture/revision	BPTB 4SSTG	24 24	45.5 ± 8.7† 42.5 ± 7.5†	9/15† 11/13†
Lu, 2017, People's Republic of China[49]	18 mo	Lysholm Tegner Pivot-shift	BPTB 4SSTG	40 59	35.2 ± 14.9# 37.5 ± 17.1#	11/29# 19/40#

continued

Table 2. (*continued*)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age *(yr)*	Sex, F/M
Webster, 2016, Australia[50]	15 yr	IKDC Instrumented laxity Rerupture/revision Return to sport	BPTB	22	26.6 ± 6.7	6/16†
			4SSTG	25	26.1 ± 5.9	5/20†
Mohtadi, 2015, Canada[52]§§	2 yr	Instrumented laxity	BPTB	106	28.7 ± 9.7	47/63
			4SSTG	108	28.5 ± 9.9	51/59
Karimi-Mobarakeh, 2015, Iran[75]##	1 yr	Return to sport	4SSTG	61	29.7 ± 7.9	11/50†
			4SST	58	28.8 ± 8.2	10/48†
Barenius, 2014, Sweden[55]***	14 yr	Lysholm Tegner Rerupture/revision	BPTB	69	39.2 ± 5.6†	34/35†
			4SST	65	41.6 ± 7.0†	21/44†
Lund, 2014, Denmark[54]	2 yr	IKDC Pivot-shift Rerupture/revision	QTB	21	30 ± 9	5/20
			BPTB	18	31 ± 8	4/22
Papalia, 2015, Italy[51]	1 yr	IKDC Lysholm	BPTB	20	Median (range): 24.7 (17 to 30)	20/0
			4SSTG	20	Median (range): 25.4 (18 to 36)	20/0
Razi, 2014, Iran[53]	3 yr	Tegner Pivot-shift	BPTB	37	30.8 ± 4.5	8/38
			4SSTG	34	28.2 ± 3.7	5/36
Wipfler, 2011, Germany[56]	8.8 yr	Lysholm Tegner Instrumented laxity Rerupture/revision	BPTB	29	29.9 (25 to 55)	12/19
			4SSTG	25	34.2 (26 to 64)	13/18
Barenius, 2010, Sweden[58]***	8 yr	Instrumented laxity Pivot-shift	BPTB	78	33.0 ± 6.3†	40/38†
			4SST	75	35.0 ± 7.5†	24/51†
Moraiti, 2010, Greece[57]	21 mo for BPTB, 25 mo for 4SSTG	IKDC Lysholm	BPTB	6	23 ± 2	0/6
			4SSTG	6	24 ± 3	0/6
Taylor, 2009, U.S.A.[59]	2.7 yr for BPTB, 3.2 yr for 4SSTG	Lysholm Tegner Rerupture/revision	BPTB	21	Median (range): 21.7 (18 to 37)	7/25
			4SSTG	24	Median (range): 22.1 (17 to 44)	4/28
Maletis, 2007, U.S.A.[60]	2 yr	Rerupture/revision	BPTB	46	27.2 (15 to 42)	15/31
			4SSTG	50	27.7 (14 to 48)	8/45
Zaffagnini, 2006, Italy[61]‡	5 yr	Pivot-shift Return to sport	BPTB	25	30.5 (22 to 47)	9/16
			4SSTG	25	31.3 (26 to 49)	10/15
Harilainen, 2006, Finland[62]	5 yr	Lysholm Tegner Pivot-shift Rerupture/revision	BPTB	37	Overall mean: females, 33; males, 29	NS
			4SSTG	37		NS
Laxdal, 2005, Sweden[63]	2 yr	Lysholm Tegner Rerupture/revision	BPTB	38	Median (range): 28 (16 to 52)	11/29
			4SSTG	38	Median (range): 26 (15 to 35)	11/28
Ibrahim, 2005, Kuwait[64]	81 mo	Lysholm Tegner Return to sport	BPTB	40	Overall: 22.3 (17 to 34)	0/40
			4SSTG	45		0/45
Aglietti, 2004, Italy[65]	2 yr	Instrumented laxity	BPTB	60	25 (16 to 39)	14/46
			4SSTG	60	25 (15 to 39)	14/46
Shaieb, 2002, U.S.A.[66]	2.5 yr	Rerupture/revision Return to sport	BPTB	31	30 (14 to 53)	7/26
			4SSTG	35	32 (14 to 48)	16/21

continued

Table 2. (*continued*)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age *(yr)*	Sex, F/M
Beard, 2001, U.K.[67]	1 yr	Lysholm Tegner	BPTB	22	NS	NS
			4SSTG	23	NS	NS
Aune, 2001, Norway[68]	2 yr	Instrumented laxity Rerupture/revision	BPTB	29	25 ± 7	16/19
			4SSTG	32	27 ± 9	16/21

*Ages are given as the mean ± standard deviation or mean (range) unless otherwise indicated. Age and sex data reflect baseline unless otherwise specified. 4SST = 4-strand semitendinosus, 4SSTG = 4-strand semitendinosus and gracilis, 5SSTG = 5-strand semitendinosus and gracilis, BPTB = bone-patellar tendon-bone, FQT = free quadriceps tendon, QTB = quadriceps tendon with bone block, IKDC = International Knee Documentation Committee Subjective Knee Form, NS = not specified, IQR = interquartile range. †Data reflect final follow-up. ‡2 reports of 1 RCT. §We had been concerned that these 2 studies might share overlapping data; however, as that could not be verified and they were conducted in different settings, we treated them as separate studies. #Whether data reflect baseline or final follow-up was not specified. **Although 4SSTG was used in <10% of cases, all cases were ultimately analyzed as 4SST based on the intention-to-treat principle. ††The sum of the 2 sexes contradicts the number of participants. ‡‡Although 5SSTG was used in some cases, all cases were ultimately analyzed as 4SSTG based on the intention-to-treat principle. §§2 reports of 1 RCT. ##This RCT was not included in the network meta-analysis. ***2 reports of 1 RCT.

gracilis (4SSTG) (SUCRA = 40.6%), FQT (SUCRA = 31.1%), and BPTB (SUCRA = 22.6%).

Lysholm Score

The Lysholm NMA included 30 studies with 1,950 patients[28-32,34-41,43,44,47-49,51,55-57,59,62-64,67,69,73,74]. The greatest difference was observed between FQT and BPTB (MD = 3.44 in favor of FQT, 95% CrI: −0.60 to 7.46). However, none of the pairwise comparisons for the Lysholm score were significant. According to the SUCRA values (Fig. 2-B), FQT was the most likely to be the best autograft (SUCRA = 87.4%) on the basis of the Lysholm score, followed by 5SSTG (SUCRA = 78.8%), 4SST (SUCRA = 49.2%), 4SSTG (SUCRA = 40.9%), and QTB (SUCRA = 32.7%), while BPTB was the least likely (SUCRA = 11.1%).

Tegner Scale

The Tegner NMA included 18 studies with 1,326 patients[29,30,32,35,36,40,43,48,49,53,55,56,59,62-64,67,72]. The comparison between QTB and 5SSTG showed the greatest difference (MD = 1.10 in favor of QTB, 95% CrI: −1.13 to 3.29). No significant differences in the Tegner scale were observed between autografts. In the autograft rankings (Fig. 2-C), QTB scored highest (SUCRA = 85.3%). The moderately performing autografts 4SST, FQT, and BTPB achieved SUCRA values of 51.9%, 49.5%, and 46.2%, respectively. In contrast, 5SSTG was the least likely to be the best autograft (SUCRA = 30.8%), followed by 4SSTG (SUCRA = 36.4%).

Instrumented AP Laxity Measurement

A total of 16 studies with 1,298 patients were incorporated into the NMA of AP laxity[32,33,35,39,40,45,46,48,50,52,56,58,65,68,69,72]. The largest difference was observed between BPTB and 5SSTG (MD = 0.57 mm in favor of BPTB, 95% CrI: −0.73 to 1.88 mm). All autografts were comparable with respect to AP laxity, with no significant differences observed. The rankings indicated that BPTB was the top-performing autograft (SUCRA = 85.7%), as shown in Figure 2-D. The remaining autografts, in order of descending rank, were QTB (SUCRA = 52.0%), 4SST (SUCRA = 47.5%), 4SSTG (SUCRA = 43.5%), and FQT (SUCRA = 39.5%), with 5SSTG showing the least relative performance (SUCRA = 31.7%).

Pivot-Shift Test

The pivot-shift NMA included 14 studies with 1,250 patients[32,33,36,40,41,46,49,53,54,58,61,62,70,72]. QTB showed a significantly lower risk of a pivot-shift grade of 2+ or higher compared with 4SST (RR = 0.26, 95% CrI: 0.07 to 0.85). Based on the rankings (Fig. 2-E), QTB demonstrated the highest probability of outperforming other autografts (SUCRA = 88.3%), followed by 5SSTG (SUCRA = 58.1%), BPTB (SUCRA = 57.4%), and 4SSTG (SUCRA = 38.5%); in contrast, 4SST ranked lowest (SUCRA = 7.7%).

Rerupture and Revision ACLR

The NMA for rerupture or revision ACLR included 30 studies with 2,462 patients[29,32-36,39-43,45,46,48,50,54-56,59,60,62,63,66,68-74]. No pairwise comparison indicated a significant difference in the risk of rerupture or revision ACLR. According to the rankings, QTB was the top-performing autograft (SUCRA = 83.3%). Autografts with intermediate performance were 4SSTG (SUCRA = 52.5%), BPTB (49.9%), FQT (45.5%), and 4SST (40.7%). With a SUCRA value of 28.1%, 5SSTG was identified as the least favorable autograft (Fig. 2-F).

Return to Sport

The qualitative synthesis of the return to sport comparisons included 14 studies[28,30,35,37,39,41,45,46,50,61,64,66,73,75] and is provided in Appendix 7. No evidence of significant differences was observed between the autografts.

Convergence, Inconsistency, Publication Bias, and Sensitivity Analysis

All outcomes showed good convergence and no inconsistency between direct and indirect evidence (see Appendices 8 and 9,

Table 3. League Table Showing the Mean Difference or Risk Ratio for Each Pairwise Comparison*

IKDC: Mean Difference (95% Credible Interval)					
QTB	−1.50 (−5.72 to 2.36)	−2.59 (−7.71 to 2.25)	−2.70 (−6.11 to 0.39)	−3.18 (−8.27 to 1.31)	−3.46 (−6.77 to −0.29)*
1.50 (−2.36 to 5.72)	4SST	−1.09 (−5.30 to 3.12)	−1.20 (−3.96 to 1.42)	−1.68 (−5.62 to 1.92)	−1.95 (−5.13 to 1.28)
2.59 (−2.25 to 7.71)	1.09 (−3.12 to 5.30)	5SSTG	−0.11 (−4.09 to 3.66)	−0.59 (−3.91 to 2.18)	−0.86 (−5.27 to 3.59)
2.70 (−0.39 to 6.11)	1.20 (−1.42 to 3.96)	0.11 (−3.66 to 4.09)	4SSTG	−0.48 (−4.14 to 2.97)	−0.76 (−3.01 to 1.65)
3.18 (−1.31 to 8.27)	1.68 (−1.92 to 5.62)	0.59 (−2.18 to 3.91)	0.48 (−2.97 to 4.14)	FQT	−0.27 (−4.27 to 4.07)
3.46 (0.29 to 6.77)*	1.95 (−1.28 to 5.13)	0.86 (−3.59 to 5.27)	0.76 (−1.65 to 3.01)	0.27 (−4.07 to 4.27)	BPTB
Lysholm: Mean Difference (95% Credible Interval)					
FQT	−0.28 (−1.84 to 1.39)	−2.55 (−6.60 to 1.61)	−2.87 (−6.82 to 1.09)	−3.03 (−7.17 to 1.04)	−3.44 (−7.46 to 0.60)
0.28 (−1.39 to 1.84)	5SSTG	−2.27 (−6.23 to 1.74)	−2.59 (−6.43 to 1.25)	−2.75 (−6.80 to 1.20)	−3.16 (−7.05 to 0.74)
2.55 (−1.61 to 6.60)	2.27 (−1.74 to 6.23)	4SST	−0.32 (−2.05 to 1.08)	−0.48 (−2.59 to 1.19)	−0.89 (−2.68 to 0.65)
2.87 (−1.09 to 6.82)	2.59 (−1.25 to 6.43)	0.32 (−1.08 to 2.05)	4SSTG	−0.17 (−1.57 to 1.09)	−0.57 (−1.35 to 0.25)
3.03 (−1.04 to 7.17)	2.75 (−1.20 to 6.80)	0.48 (−1.19 to 2.59)	0.17 (−1.09 to 1.57)	QTB	−0.41 (−1.67 to 1.06)
3.44 (−0.60 to 7.46)	3.16 (−0.74 to 7.05)	0.89 (−0.65 to 2.68)	0.57 (−0.25 to 1.35)	0.41 (−1.06 to 1.67)	BPTB
Tegner: Mean Difference (95% Credible Interval)					
QTB	−0.60 (−1.88 to 0.70)	−0.66 (−2.72 to 1.43)	−0.71 (−1.87 to 0.44)	−0.81 (−1.95 to 0.35)	−1.10 (−3.29 to 1.13)
0.60 (−0.70 to 1.88)	4SST	−0.05 (−2.11 to 2.02)	−0.11 (−1.25 to 1.01)	−0.21 (−1.33 to 0.91)	−0.49 (−2.69 to 1.70)
0.66 (−1.43 to 2.72)	0.05 (−2.02 to 2.11)	FQT	−0.05 (−1.88 to 1.75)	−0.15 (−1.89 to 1.57)	−0.44 (−3.01 to 2.11)
0.71 (−0.44 to 1.87)	0.11 (−1.01 to 1.25)	0.05 (−1.75 to 1.88)	BPTB	−0.10 (−0.65 to 0.46)	−0.38 (−2.35 to 1.58)
0.81 (−0.35 to 1.95)	0.21 (−0.91 to 1.33)	0.15 (−1.57 to 1.89)	0.10 (−0.46 to 0.65)	4SSTG	−0.28 (−2.15 to 1.60)
1.10 (−1.13 to 3.29)	0.49 (−1.70 to 2.69)	0.44 (−2.11 to 3.01)	0.38 (−1.58 to 2.35)	0.28 (−1.60 to 2.15)	5SSTG
Instrumented Anteroposterior Laxity in mm: Mean Difference (95% Credible Interval)					
BPTB	0.26 (−0.38 to 0.92)	0.30 (−0.23 to 0.85)	0.31 (−0.07 to 0.68)	0.40 (−0.59 to 1.40)	0.57 (−0.73 to 1.88)
−0.26 (−0.92 to 0.38)	QTB	0.04 (−0.79 to 0.85)	0.05 (−0.57 to 0.64)	0.14 (−1.04 to 1.30)	0.31 (−1.11 to 1.68)
−0.30 (−0.85 to 0.23)	−0.04 (−0.85 to 0.79)	4SST	0.01 (−0.60 to 0.60)	0.10 (−0.74 to 0.96)	0.27 (−1.12 to 1.64)
−0.31 (−0.68 to 0.07)	−0.05 (−0.64 to 0.57)	−0.01 (−0.60 to 0.60)	4SSTG	0.08 (−0.93 to 1.14)	0.26 (−0.99 to 1.51)
−0.40 (−1.40 to 0.59)	−0.14 (−1.30 to 1.04)	−0.10 (−0.96 to 0.74)	−0.08 (−1.14 to 0.93)	FQT	0.17 (−1.43 to 1.77)
−0.57 (−1.88 to 0.73)	−0.31 (−1.68 to 1.11)	−0.27 (−1.64 to 1.12)	−0.26 (−1.51 to 0.99)	−0.17 (−1.77 to 1.43)	5SSTG
Pivot-Shift Grade ≥2+: Risk Ratio (95% Credible Interval)					
QTB	1.58 (0.22 to 11.23)	1.73 (0.71 to 4.62)	2.10 (0.90 to 5.64)	3.81 (1.17 to 14.78)*	No data
0.63 (0.09 to 4.59)	5SSTG	1.10 (0.18 to 6.96)	1.33 (0.24 to 8.00)	2.40 (0.35 to 18.20)	No data
0.58 (0.22 to 1.41)	0.91 (0.14 to 5.43)	BPTB	1.20 (0.74 to 2.07)	2.17 (0.86 to 6.22)	No data
0.48 (0.18 to 1.12)	0.75 (0.12 to 4.19)	0.83 (0.48 to 1.35)	4SSTG	1.80 (0.77 to 4.62)	No data
0.26 (0.07 to 0.85)*	0.42 (0.05 to 2.82)	0.46 (0.16 to 1.16)	0.56 (0.22 to 1.30)	4SST	No data
No data	No data	No data	No data	No data	FQT
Rerupture/Revision ACLR: Risk Ratio (95% Credible Interval)					
QTB	1.73 (0.64 to 5.09)	1.77 (0.65 to 5.41)	2.18 (0.04 to 107.88)	2.04 (0.57 to 8.01)	2.86 (0.63 to 17.10)
0.58 (0.20 to 1.57)	4SSTG	1.02 (0.69 to 1.54)	1.26 (0.03 to 51.17)	1.17 (0.51 to 2.77)	1.66 (0.35 to 10.02)
0.57 (0.18 to 1.54)	0.98 (0.65 to 1.45)	BPTB	1.24 (0.02 to 51.13)	1.15 (0.47 to 2.85)	1.62 (0.33 to 10.13)
0.46 (0.01 to 24.94)	0.80 (0.02 to 38.65)	0.81 (0.02 to 41.16)	FQT	0.94 (0.02 to 42.89)	1.37 (0.02 to 91.48)
0.49 (0.12 to 1.74)	0.85 (0.36 to 1.96)	0.87 (0.35 to 2.12)	1.06 (0.02 to 41.92)	4SST	1.43 (0.24 to 9.57)
0.35 (0.06 to 1.59)	0.60 (0.10 to 2.88)	0.62 (0.10 to 3.03)	0.73 (0.01 to 44.76)	0.70 (0.10 to 4.18)	5SSTG

*A positive value indicates that the graft named in the column is superior to the graft named in the row by the stated amount. 4SST = 4-strand semitendinosus, 4SSTG = 4-strand semitendinosus and gracilis, 5SSTG = 5-strand semitendinosus and gracilis, BPTB = bone-patellar tendon-bone, FQT = free quadriceps tendon, QTB = quadriceps tendon with bone block, IKDC = International Knee Documentation Committee Subjective Knee Form.

including Supplementary Figs. 4-A through 5-E). No publication bias was detected except in the pivot-shift outcome, where the Egger test showed significant bias (see Appendix 10, including Supplementary Figs. 6-A through 6-E). After excluding 3 outlier studies[32,61,62] in a sensitivity analysis, SUCRA rankings for the pivot-shift remained unchanged, supporting the robustness of the primary findings.

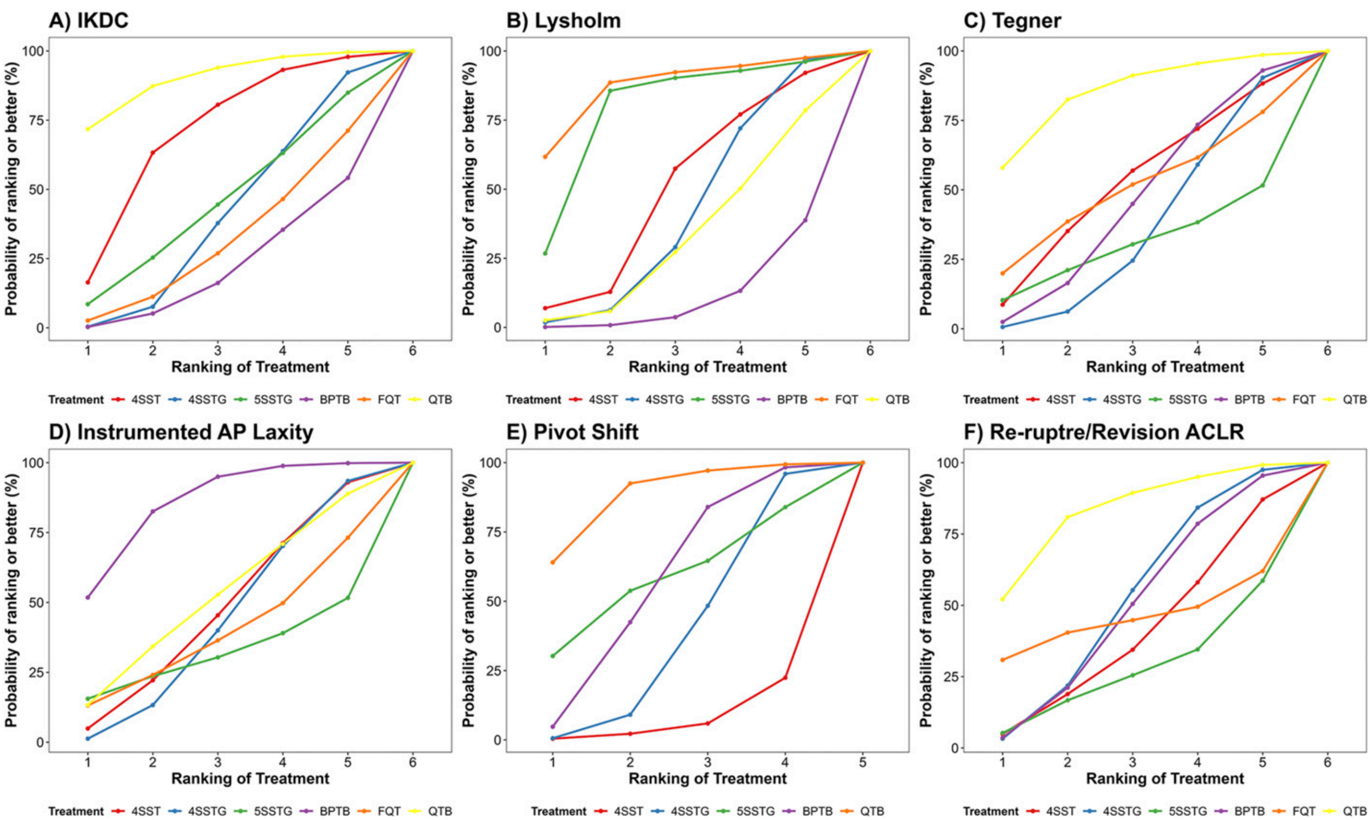

Fig. 2

SUCRA plots illustrating the ranking probabilities of the 6 autografts for each analyzed outcome: IKDC score, Lysholm score, Tegner scale, instrumented anteroposterior laxity, pivot-shift, and rerupture or revision risk.

Network Meta-Regression

The interaction between autograft type and follow-up time for rerupture or revision ACLR risk revealed no significant evidence that the RR of any comparison changed over time (see Appendix 11, including Supplementary Fig. 7).

Discussion

This study represents the most comprehensive evidence synthesis to date on autograft selection for primary ACLR. Based on our findings, QTB was identified as the most favorable autograft option overall, consistently ranking first or second in all outcomes except the Lysholm score. In contrast, BPTB was the best autograft option with respect to AP laxity but was among the least favorable options with respect to PROMs, ranking the lowest for both the IKDC and Lysholm scores. Also, considerable performance variability was observed among the HT subtypes, with no single variant emerging as consistently superior.

QT autografts performed the best overall with respect to the PROMs. Specifically, QTB was the most favorable autograft according to the IKDC score and Tegner scale, whereas FQT achieved the highest Lysholm scores. QT autografts minimize bone-block harvest burden, avoid saphenous nerve injury inherent to patellar tendon harvesting[76], and preserve superior histological integrity of the graft-tendon interface[77], resulting in less anterior knee pain and preserved kneeling tolerance, which translate to superior PROMs[78]. BPTB was the least favorable autograft with respect to both the IKDC and Lysholm scores. BPTB harvesting may directly damage the extensor mechanism, resulting in increased anterior knee symptoms, kneeling pain, and early extensor weakness or extensor lag[79], which weigh heavily on PROMs. Therefore, although BPTB demonstrates mechanical superiority, it results in comparatively worse PROMs. In the present study, a significant difference was observed between QTB and BPTB, with an MD of 3.5 in favor of QTB for the IKDC score. However, this MD falls below the established MCID, indicating that it is unlikely to be clinically meaningful[80,81]. Previous NMAs have reported conflicting PROM findings for QT autografts, likely because some did not distinguish between QTB and FQT variants[13,17]. In the current study, we could not identify any RCT directly comparing QTB and FQT autografts; however, our NMA indicated that, unlike QTB, FQT was not a top-performing autograft with respect to the IKDC score and Tegner scale. Therefore, future RCTs directly comparing FQT and QTB autografts for primary ACLR are encouraged.

With respect to knee stability, 4SST use was associated with a fourfold higher risk of a grade 2+ or higher pivot-shift, which was significant, compared with QTB, which was the best autograft option for this outcome. A previous meta-analysis reported that a grade 2+ or higher pivot-shift was less frequently observed for QT compared with HT[82]. In addition, an NMA comparing a heterogeneous set of autografts, allografts, and synthetic grafts also revealed that QT achieved the highest SUCRA value compared with all other graft types based on the pivot-shift[15]. Our findings revealed that BPTB ranked best for instrumented AP laxity (SUCRA = 86%, highest MD = 0.6 mm). BPTB autograft may have demonstrated nonsignificantly superior stability primarily through bone-to-bone healing, which occurs more reliably than tendon-to-bone healing seen with soft-tissue autografts[83]. The bone plugs undergo direct osseous integration into the bone tunnels, providing immediate mechanical fixation that may prevent graft slippage[7]. Likewise, a previous NMA comparing a heterogeneous set of grafts reported BPTB to be the most favorable option with respect to instrumented KT-1000/2000 (MEDmetric) results, with lower performance regarding all other knee stability outcomes. Importantly, our analysis demonstrated that QTB ranked as the first and second most favorable autograft for pivot-shift and AP laxity, respectively, whereas that previous NMA did not include any subtype of QT[18].

In our NMA, QTB was the most favorable autograft with respect to the risk of ACL rerupture or revision ACLR. This interpretation, however, should be made with caution, as none of the comparisons reached significance. Likewise, a previous NMA noted that QTB use was associated with the lowest risk of rerupture[17]. Also, our systematic review revealed that return-to-sport comparisons remain limited by heterogeneity in study methods and insufficient evidence regarding newer autografts such as FQT.

This study has some limitations. Donor-site morbidity and complication rates were not included as evaluated outcomes. In addition, the included studies varied with respect to autograft preparation, fixation methods, and rehabilitation protocols, with many lacking clear reporting of these details. Furthermore, the follow-up durations varied considerably among studies, ranging between 1 and 20 years. To account for this, however, we performed a meta-regression on follow-up duration for the ACL rerupture outcome.

In summary, this study indicated that QTB consistently outperformed other autografts, showing more favorable results for most outcomes compared with BPTB and the HT subtypes. Our findings suggest that using QTB autograft may lead to improved functional, activity-related, and stability outcomes, while also reducing the risk of graft failure. BPTB performed well with respect to AP knee laxity measures but poorly with respect to PROMs, whereas comparisons among HT subtypes were inconsistent and showed no clear superiority relative to each other in any specific domain. Given the subtle and mainly nonsignificant differences among the autografts, autograft selection should remain a multifaceted decision that balances the favorable profile of QTB against the risks and benefits associated with each alternative autograft option, tailored to patient-specific goals. Future RCTs focusing more on return to sport outcomes and directly comparing FQT outcomes with QTB are encouraged.

Appendix

eA Supporting material provided by the authors is posted with the online version of this article as a data supplement at jbjs.org (http://links.lww.com/JBJS/J136). ■

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